

CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

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Case 16-2023: A 13-Year-Old Boy with Depression and Hypotension

Kevin R. Schwartz, M.D., Abigail L. Donovan, M.D., Bryan D. Hayes, Pharm.D.,
Mai Uchida, M.D., and Joy B. Rosen, M.S.

PRESENTATION OF CASE

From the Departments of Pediatrics (K.R.S.), Emergency Medicine (K.R.S., B.D.H.), Psychiatry (A.L.D., M.U.), Pharmacy (B.D.H.), and Behavioral Health (J.B.R.), Massachusetts General Hospital, and the Departments of Pediatrics (K.R.S.), Emergency Medicine (K.R.S., B.D.H.), Psychiatry (A.L.D., M.U.), and Behavioral Health (J.B.R.), Harvard Medical School — both in Boston.

N Engl J Med 2023;388:1992-9.

DOI: 10.1056/NEJMcpc2201237

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Dr. Abigail L. Donovan: A 13-year-old boy with a history of attention deficit-hyperactivity disorder (ADHD), borderline personality disorder, and major depressive disorder was evaluated in the emergency department of this hospital because of worsening depression, thoughts of self-injurious behavior, and suicidality.

The patient had a history of impulsivity and frequent episodes of self-injury, such as head-butting and cutting, which sometimes occurred while he was in treatment settings. Twelve months before the current presentation, he began to have suicidal ideation, with occasional suicidal planning and a suicide attempt. During the next year, the patient had 10 evaluations in the emergency department and six inpatient psychiatric hospitalizations.

One day before the current presentation, the patient was discharged from an inpatient psychiatric hospital, where he had been admitted and treated for 3 weeks because of urges to hurt himself. On the day of the current presentation, the patient visited his middle school and participated in therapy at home. Throughout the day, he noticed worsening depression and had recurrent thoughts of self-injury and suicide. He asked his parents to call emergency medical services because he felt unsafe in his home. He was brought to the emergency department of this hospital.

On arrival, the patient was in an agitated and combative state, reporting that he had changed his mind and no longer wanted to be in the hospital. Attempts at verbal deescalation were not effective. Treatment with intramuscular olanzapine was administered, and physical restraints were placed on his arms and legs. During the next 6 hours, the agitation decreased, and the physical restraints were removed. Additional history was obtained but was limited because the patient shrugged his shoulders in response to most questions. He described the school visit as difficult and uncomfortable. He did not indicate why he felt unsafe in his home; he reported that he had picked at scabs on his arms and legs in order to hurt himself and had ongoing suicidal ideation. The patient also reported difficulty sleeping and concentrating, as well as loss of interest and motivation,

although he still enjoyed playing sports. He had no homicidal ideation, paranoid thoughts, or delusions.

Medications included extended-release guanfacine and lurasidone. The patient had no known drug allergies. He did not smoke cigarettes, drink alcohol, or use illicit drugs. He lived in an urban area of New Hampshire with his mother, father, and sister.

On examination, the temporal temperature was 36.4°C, the blood pressure 133/78 mm Hg, the heart rate 76 beats per minute, the respiratory rate 18 breaths per minute, and the oxygen saturation 98% while the patient was breathing ambient air. He appeared disheveled. There were scattered cuts and wounds in varying stages of healing on both arms. The patient was in a disengaged and agitated state. He had a depressed mood, a guarded affect, and difficulty concentrating. His speech was slow and soft, with reduced prosody. His thought process was logical and linear; the thought content was notable for suicidal ideation. He had no loosening of associations, delusions, or hallucinations. Insight and judgment were poor.

The patient was deemed to be at high risk for harm and was transferred to the acute psychiatric service of the emergency department. Transfer to an inpatient psychiatric hospital was requested, but no beds were available at any facility in the region. During the next 6 weeks, while the patient was awaiting admission to an inpatient psychiatric hospital, treatment with extended-release guanfacine and lurasidone was continued, and treatment with sertraline, diphenhydramine, and melatonin was initiated. He had frequent episodes of agitation, which led to the administration of oral clonidine or oral or intramuscular olanzapine as needed. He also had several episodes of self-injury; surveillance was heightened, with well-being checks performed every 15 minutes.

On day 41 in the acute psychiatric service of the emergency department, agitation developed, and a dose of oral olanzapine was administered in addition to scheduled medications. One hour later, agitation continued, and a dose of intramuscular olanzapine was administered. After the administration of intramuscular olanzapine, the temporal temperature was 36.4°C, the blood pressure 119/65 mm Hg, the heart rate 73 beats per minute, the respiratory rate 18 breaths per

minute, and the oxygen saturation 98% while the patient was breathing ambient air.

The next morning, the blood pressure was 146/87 mm Hg and the heart rate 52 beats per minute. There were no new symptoms; the examination was unchanged. Twelve hours later, the blood pressure was 69/39 mm Hg and the heart rate 88 beats per minute. The patient reported malaise and lightheadedness with standing, as well as nausea and abdominal pain. He appeared pale, tired, and lethargic. Oral rehydration was encouraged. One hour later, the blood pressure was 94/50 mm Hg and the heart rate 88 beats per minute. Three hours after the onset of hypotension, the blood pressure was 79/45 mm Hg and the heart rate 85 beats per minute.

Blood and urine toxicologic screening was negative. Laboratory test results are shown in Table 1. A blood specimen was obtained for culture. Intravenous fluids, vancomycin, and ceftriaxone were administered. Computed tomography (CT) of the abdomen and pelvis, performed after the administration of intravenous contrast material, revealed no abnormalities. The patient was admitted to the pediatric medical service of this hospital.

A diagnosis was made.

DIFFERENTIAL DIAGNOSIS

Dr. Kevin R. Schwartz: This 13-year-old boy with ADHD, borderline personality disorder, and major depressive disorder had a prolonged stay in the acute psychiatric service of the emergency department while he was awaiting admission to an inpatient psychiatric hospital. There was a history of suicidality, several episodes of self-injury, and treatment with multiple psychiatric medications. On day 41 in the emergency department, transient hypertension and bradycardia developed, followed by profound hypotension and lethargy.

If I were the emergency medicine physician asked to evaluate this patient, I would begin with rapid assessment and stabilization of the airway, breathing, and circulation. In this case, the airway was patent and the breathing unlabored, but the circulation was markedly abnormal, with hypotension despite a normal heart rate for his age. The differential diagnosis for this patient's hypotension should begin with consideration of the most life-threatening possibility: decompensated shock.

Table 1. Laboratory Data.*

Variable	Reference Range†	Hospital Day 42
Hemoglobin (g/dl)	12.0–16.0	12.8
Hematocrit (%)	36.0–46.0	39.3
White-cell count (per μ l)	4500–13,500	4390
Platelet count (per μ l)	150,000–400,000	367,000
Sodium (mmol/liter)	135–145	138
Potassium (mmol/liter)	3.4–5.0	3.7
Chloride (mmol/liter)	98–108	104
Carbon dioxide (mmol/liter)	23–32	26
Urea nitrogen (mg/dl)	8–25	6
Creatinine (mg/dl)	0.60–1.50	0.76
Glucose (mg/dl)	70–110	132
Calcium (mg/dl)	8.5–10.5	9.3
Magnesium (mg/dl)	1.7–2.4	2.0
C-reactive protein (mg/liter)	0.0–8.0	32.3
Lactic acid (mmol/liter)	0.5–2.0	1.0
Alanine aminotransferase (U/liter)	7–33	15
Aspartate aminotransferase (U/liter)	9–32	16
Alkaline phosphatase (U/liter)	57–254	106
Total bilirubin (mg/dl)	0.0–1.0	0.5

* To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for glucose to millimoles per liter, multiply by 0.05551. To convert the values for calcium to millimoles per liter, multiply by 0.250. To convert the values for magnesium to millimoles per liter, multiply by 0.4114. To convert the values for lactic acid to milligrams per deciliter, divide by 0.1110. To convert the values for bilirubin to micromoles per liter, multiply by 17.1.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

SHOCK

Shock is defined as the failure of oxygen delivery to meet the metabolic demands of tissue. It is typically a consequence of inadequate perfusion. In children with shock, the blood pressure can remain normal because of compensatory mechanisms that increase the heart rate, stroke volume, peripheral vascular resistance, and venous tone. Therefore, in such patients, hypotension is considered to be a late and ominous finding that is indicative of decompensated shock.¹ The next step in evaluating this patient is to consider possible causes of shock so that an appropriate treatment can be promptly initiated.¹

DISTRIBUTIVE SHOCK

Distributive shock occurs when vasodilation results in the inappropriate distribution of blood flow. Among the causes of distributive shock, sepsis is the most likely possibility in this patient. Consensus definitions of pediatric sepsis have been established.² However, when criteria that are less specific than the established definitions are used to identify pediatric patients who are at risk for sepsis, and when empirical treatment with intravenous fluids and broad-spectrum antibiotic therapy is administered rapidly, survival is higher.^{3,4} In this patient with hypotension and lethargy, sepsis was appropriately suspected, blood cultures were obtained, and treatment with intravenous fluids and broad-spectrum antibiotics was initiated. However, the patient had no fever, tachycardia, abnormal white-cell count, or symptoms that would be suggestive of infection, so other causes of shock must be considered.

HYPOVOLEMIC SHOCK

Hypovolemic shock, the most common cause of shock in children worldwide, is caused by depletion of intravascular volume. Could this patient have hypovolemic shock due to severe dehydration that developed during his prolonged stay in the acute psychiatric service of the emergency department? When hypotension occurs with dehydration, the deficit of total body water is typically greater than 10%. This degree of volume depletion is usually accompanied by other findings, such as tachycardia, dry mucosa, and delayed capillary refill, none of which were present in this patient.⁵

Could this patient have hypovolemic shock due to hemorrhage? When hypotension occurs with hemorrhage, the loss of blood volume is typically greater than 30%. This degree of blood loss can occur in six situations: internal bleeding in the chest, abdomen, pelvis, or retroperitoneum or around the femurs or external bleeding. The patient's abdominal pain suggests the possibility of blunt abdominal trauma that has led to bleeding in the abdomen. However, CT of the abdomen, the preferred assessment in pediatric patients with blunt abdominal trauma, ruled out intraabdominal injury.⁶

CARDIOGENIC SHOCK

Cardiogenic shock is caused by inadequate cardiac output. In children, cardiogenic shock is

most commonly due to myocarditis. Myocarditis is typically caused by a viral infection and may be accompanied by a prodrome, chest pain, respiratory distress, hepatomegaly, or tachycardia. Electrocardiographic changes are often present but can be nonspecific; troponin levels may be elevated.⁷

Lyme carditis could be considered in this patient, given that he is from a region where Lyme disease is endemic. Lyme carditis occurs during the early disseminated phase of the disease (weeks to months after the initial infection) and is most frequently associated with atrioventricular block, which can be of a high grade. It can also be associated with myopericarditis that leads to a decreased left ventricular ejection fraction.⁸ Could a combination of atrioventricular block and a decreased left ventricular ejection fraction explain this patient's hypotension without tachycardia? Although this possibility is unlikely, I would request that an electrocardiogram and troponin levels be obtained before considering other causes.

OBSTRUCTIVE SHOCK

Obstruction of arterial outflow or venous inflow to the heart results in insufficient circulation and obstructive shock. Causes include cardiac tamponade, pulmonary embolism, and tension pneumothorax. These conditions are unlikely in this patient in the absence of accompanying tachycardia and chest pain.

MEDICATION EFFECT

This patient had an episode of hypertension and bradycardia 12 hours before hypotension developed. Could there be a unifying explanation for both the transient hypertension with bradycardia and the subsequent persistent hypotension? Careful review of the patient's medications, with examination of side effects and possible drug interactions, would be helpful.

Sertraline, diphenhydramine, melatonin, and lurasidone alone are not known to cause hypotension, although lurasidone in combination with guanfacine has the potential to do so. Olanzapine has been associated with hypotension but not with relative bradycardia.⁹ Guanfacine and clonidine are both centrally acting α_2 -adrenergic agonists that were initially developed as antihypertensive medications but are increasingly being

used for the treatment of behavioral disorders, especially ADHD. Both medications have a narrow therapeutic window, and an overdose can cause drowsiness, bradycardia, and hypotension.^{10,11} An overdose can also result in initial peripherally acting α_2 -adrenergic agonism, and both medications have been associated with transient hypertension that precedes sustained hypotension, which was seen in this patient.^{12,13}

I think that this patient's hypotension — which occurred in the absence of tachycardia, fever, and clinically significant laboratory abnormalities and was preceded by transient hypertension with bradycardia — can best be explained by an ingestion of guanfacine or clonidine that led to shock from decreased sympathetic outflow. I would ask the patient whether he had ingested additional doses of either of these medications.

DR. KEVIN R. SCHWARTZ'S DIAGNOSIS

Toxic effects of a centrally acting α_2 -adrenergic agonist, either guanfacine or clonidine.

HOSPITAL COURSE AND CLINICAL IMPRESSION

Dr. Donovan: Guanfacine overdose was suspected, and an investigation was initiated. Because the extended-release formulation of guanfacine was not on the hospital formulary, the patient's family had brought in his home supply. The bottle of guanfacine was securely stored during the patient's extended stay in the acute psychiatric service of the emergency department. However, hospital policy required that the entire bottle be brought into the patient's room each time that the medication was dispensed. On hospital day 41, the bottle of guanfacine was brought into the patient's room in preparation for dispensing. However, the staff member was then urgently called out of the room, and the bottle of guanfacine was accidentally left behind. When directly questioned, the patient reported that he had ingested guanfacine from the bottle left in his room.

CLINICAL DIAGNOSIS

Intentional guanfacine overdose.

DISCUSSION OF MANAGEMENT

Dr. Bryan D. Hayes: Guanfacine is an imidazoline derivative, which is a class of centrally acting antihypertensive medications that also includes clonidine and dexmedetomidine. Guanfacine can cause clinically significant toxic effects, particularly in children. From 2000 to 2016, more than 11,000 cases of guanfacine exposure in children were reported to poison centers in the United States.^{10,14} There was a sharp increase after 2009, particularly among children 6 to 12 years of age.

Guanfacine is more selective for α_2 -adrenergic receptors than clonidine (by a factor of 12 to 25).¹⁵ Stimulation of α_2 -adrenergic receptors ultimately reduces sympathetic outflow, thereby decreasing norepinephrine release. Imidazoline receptors, which function upstream from presynaptic α_2 -adrenergic receptors, are also stimulated. Stimulation of the I_2 receptor mediates the release of β -endorphins that act on opioid receptors, producing opioid-like effects.

Guanfacine is readily absorbed. The peak concentration occurs 3 to 5 hours after the ingestion of an immediate-release dose. The volume of distribution is large, with 70% protein binding. Half the drug is metabolized in the liver by cytochrome P450 3A4, and the other half is excreted from the kidneys in its parent compound. The half-life is reported to be 17 hours, but a more important metric in treating toxic effects is the duration of action, which is at least 24 hours and can be longer with an extended-release formulation.

The onset of toxic effects is expected to occur 30 to 90 minutes after the ingestion of an immediate-release formulation but can be delayed with the use of an extended-release formulation. Hypotension and bradycardia are the primary, and most worrisome, effects of poisoning. Depression of the central nervous system (CNS) can have manifestations ranging from lethargy to coma with a need for airway support. Miosis is common, and hypothermia has been reported.

Treatment is largely supportive, with the goal of maintaining ventilation and perfusion until the toxic effects resolve. The use of activated charcoal can be considered in patients with a recent ingestion and a patent airway. Fluids are administered for the treatment of hypotension. In addition, a vasopressor such as norepineph-

rine may be used as a first-line agent for hypotension because it has strong α -adrenergic agonist properties. Atropine can be considered for the treatment of symptomatic bradycardia; temporary transcutaneous pacing is occasionally used for the treatment of refractory bradycardia.

Because an overdose of guanfacine (and other imidazoline derivatives) produces opioid-like effects, naloxone can be used in an attempt to reverse the effects.^{12,13} Naloxone may lead to a beneficial response resulting from alteration of the CNS sympathetic outflow by endogenous opioids. High doses are used, additional doses may be necessary, and the treatment is not always effective.

Dr. Donovan: In this patient, the blood pressure increased with the administration of intravenous fluids. An electrocardiogram showed normal sinus rhythm (Fig. 1). He remained afebrile, and the blood cultures were negative; treatment with antibiotics was stopped after 48 hours. By hospital day 44, the abdominal pain had resolved. The patient continued to have episodes of agitation and self-injury. On hospital day 47, he was transferred to an inpatient psychiatric hospital.

The patient remained at the inpatient psychiatric hospital for 48 days. He was then transferred to a long-term residential treatment program, where he was treated for 1 year before returning home to live with his family.

Because of this patient's case, the hospital policy regarding the administration of home medications was revised. The revised policy requires that a single dose of the medication be removed from the securely stored bottle, placed in a labeled container, and brought to the patient.

MENTAL HEALTH CRISIS IN CHILDREN

Dr. Mai Uchida: The coronavirus disease 2019 (Covid-19) pandemic has led to substantial mental health challenges for children in the United States. The mental health crisis in children was declared a national emergency in a joint statement issued in late 2021 by the American Academy of Pediatrics, the American Academy of Child and Adolescent Psychiatry, and the Children's Hospital Association.¹⁶ An advisory regarding children's mental health was issued by the Surgeon General.¹⁷

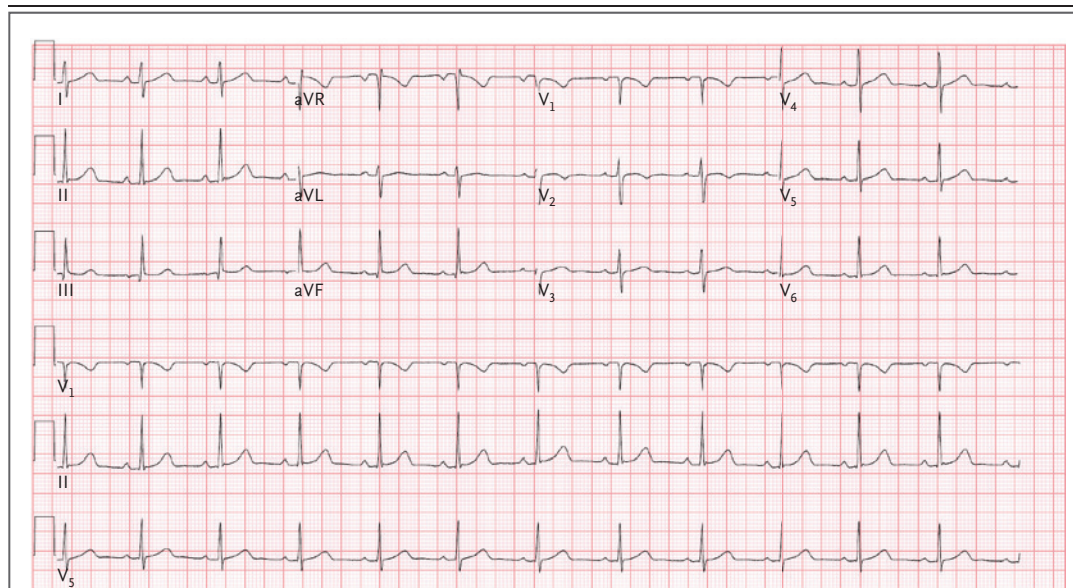


Figure 1. Electrocardiogram.

An electrocardiogram obtained after the development of hypotension shows sinus rhythm at a rate of 73 beats per minute.

In 2021, emergency department visits for suspected suicide attempts increased by 51% among girls 12 to 17 years of age, and emergency department visits for mental health emergencies increased by 24% among children 5 to 11 years of age and by 31% among children 12 to 17 years of age.¹⁸ Reports of behavioral dysregulation at school and negative interactions online have increased.¹⁹ National reading and mathematics test scores have markedly declined.²⁰

This patient presented during the Covid-19 pandemic. We cannot be certain whether his course of illness was related to the pandemic, but it is noteworthy that a large population of children in the United States started to have mental health difficulties during the pandemic, and those who already had mental health difficulties reported exacerbated symptoms. The increase in the volume of children needing higher levels of psychiatric care has led to a decrease in the availability of resources, as evidenced by this patient's experience of waiting more than 40 days for admission to an inpatient psychiatric hospital.

Although this patient did not have a history of mania, his presentation was notable for several clinical risk factors that are predictive of bipolar disorder in children presenting with

depression, including frequent episodes of irritability and agitation, severe symptoms and functional impairment (e.g., suicidality and hospitalization), and an early-onset mood dysregulation.²¹ Given the possibility of bipolar disorder, the patient was appropriately treated with mood stabilizers, including antipsychotic medications, along with antidepressant medications.

ACCESS TO PSYCHIATRIC CARE

Ms. Joy B. Rosen: This case provides a stark illustration of the crisis in Massachusetts regarding psychiatric boarding, with patients waiting in an emergency department or medical-surgical floor for a bed to become available in an inpatient psychiatric hospital. The increasing need for psychiatric boarding was a major issue before the pandemic, but the Covid-19 crisis worsened the situation by exacerbating the behavioral health crisis. According to the Massachusetts Health and Hospital Association, as of April 18, 2023, there were 515 patients in psychiatric boarding statewide, of whom 52 were children.²² Challenges in the mental health system and in emergency departments and inpatient psychiatric units have contributed to the problem regarding psychiatric boarding.

The mental health system is fragmented and difficult to access. Diversion options — such as crisis stabilization units, partial hospitalization, and easy access to behavioral health urgent care — are lacking. In addition, access to ambulatory care is limited, such that acute episodes that lead to emergency department admissions cannot be prevented. Emergency departments often do not have behavioral health providers available 24 hours a day, 7 days a week, which causes delays in evaluation and disposition planning. Inpatient psychiatric units across the state are difficult to access because a high number of licensed psychiatric beds are “blocked,” or not available to use. Blocked beds have increased because of workforce shortages, infection-control issues, and the lack of appropriate disposition placements available to patients after inpatient hospitalization.

Solutions to these problems must include enhanced access to the continuum of care services; virtual hospitalization options for patients who are residing in a structured setting (e.g., at home with family or in a residential program); telehealth options for emergency departments that are not staffed with behavioral health providers 24 hours a day, 7 days a week; and alternative treatment settings for medically and behaviorally complex pediatric patients.

Dr. Theodore A. Stern (Psychiatry): During the 6 weeks that the patient was in the acute psychi-

atric service of the emergency department, were any psychological or educational interventions applied that might have allowed for a safe discharge, rather than an inpatient psychiatric hospitalization?

Dr. Donovan: The patient was evaluated every day by a child psychiatrist, but he did not engage in therapeutic discourse, providing minimal responses to basic questions. On multiple occasions, he was offered participation in a virtual partial hospitalization program, which he declined. He was evaluated by an occupational therapist but declined to participate in the recommended interventions. The patient occasionally participated in child life and music therapy. Consistent reports of suicidal ideation and ongoing episodes of self-injury in the emergency department, coupled with limited engagement in treatment and complex family factors not discussed in the case presentation, precluded a safe discharge from the emergency department.

FINAL DIAGNOSIS

Intentional guanfacine overdose.

In this Case Record, some variables that do not affect the differential diagnosis have been changed in order to protect the patient's anonymity.

This case was presented at Psychiatry Grand Rounds.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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